LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

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EXP -12

12. Use Figma to create an interactive diagram illustrating the evolution from 1G to 5G.

Aim: -

Design an interactive diagram illustrating the evolution from 1G to 5G using Figma.

Procedure: -

1. openFigma

2. Create a new file

3. Select the Frames

4. Fill in the content that is required for presentation

5. Design Visual Elements

6. Make it Interactive

7. Add Annotations and Explanations

8. Incorporate Multimedia

9. Storyboard Animation

11. Review and edit the Prototype

12. Save and Share

Design:

Evolution of Mobile Networks · Timeline

SET 12 / MOBILE NETWORKS

01 / 07 · ACADEMIC OVERVIEW

OVERVIEW / FIVE GENERATIONS OF CELLULAR SYSTEMS

Evolution of Mobile Networks: 1G to 5G

From analog calls to programmable mobile networks — an academic overview of technologies, capabilities and trade-offs.

Five generations. One continuous evolution.

ALL DATES APPROXIMATE · ROLLOUT VARIED BY COUNTRY

1G

Late 1970s–1980s

Analog voice

AMPS · NMT · TACS

Cellular calling becomes mobile. Circuit-switched analog voice; no native packet data.

View 1G detail →

2G

Early 1990s

Digital + SMS

GSM · IS-95

Digital voice and messaging. Later GPRS / EDGE enhancements add packet data.

View 2G detail →

3G

Early 2000s

Mobile internet

UMTS / WCDMA
CDMA2000

Packet data brings web access and multimedia alongside circuit-switched voice.

View 3G detail →

4G

Late 2000s–early 2010s

IP broadband

LTE · LTE-Advanced

All-IP radio network for broadband. VoLTE where supported; legacy voice fallback otherwise.

View 4G detail →

5G

2019 onward

Connected systems

5G NR · NSA / SA

Enhanced broadband, lower-latency capabilities and massive-connectivity goals.

View 5G detail →

Evolution, not a clean replacement

Generations coexist. Voice may depend on older networks, and user experience depends on coverage, spectrum, devices and operator deployment.

Read each generation through three lenses

Technology → network architecture → voice and data. Then compare advantages, limitations and representative applications.

Not a speed ranking: theoretical peaks are not typical speeds. 5G does not universally guarantee ultra-low latency. Detail and navigation controls are static design elements.

Evolution of Mobile Networks · 1G

SET 12 / MOBILE NETWORKS

02 / 07 · ACADEMIC OVERVIEW

1G / LATE 1970S–1980S · APPROXIMATE

1G — The beginning of cellular voice

Analog radio systems introduced mobile telephony through geographically divided cells and frequency reuse.

1G

CORE TECHNOLOGIES

AMPS · NMT · TACS

VOICE
Circuit-switched analog voice. Calls use a dedicated radio channel for their duration.

DATA
No native packet data. Designed primarily for voice, not mobile internet.

How the network connects

SIMPLIFIED LOGICAL ARCHITECTURE

Voice path: Analog handset → Cell base station → Mobile switching center → Telephone network

The switching center connects calls and coordinates mobility between cells. Separate regional standards limited interoperability.

Advantages

Mobile calling beyond fixed lines; cellular frequency reuse expands geographic service and mobility.

Limitations

Limited capacity, weak security and easy eavesdropping; inconsistent quality and restricted roaming.

Representative use cases

Early car phones, business mobile calls and basic personal voice communication.

Dates are approximate: commercial rollout differed by country. Capabilities depend on deployment, spectrum, device and coverage; peak rates are not typical user speeds. Navigation controls shown are static.

← Previous · Timeline

Back to timeline

Next · 2G →

Evolution of Mobile Networks · 2G

SET 12 / MOBILE NETWORKS

03 / 07 · ACADEMIC OVERVIEW

2G / EARLY 1990S · APPROXIMATE

2G — Digital calls and text messaging

Digital radio improves spectrum use and introduces mass-market SMS; packet data arrives through later enhancements.

2G

CORE TECHNOLOGIES

GSM · IS-95
Later: GPRS / EDGE (GSM)

VOICE
Digital circuit-switched voice, with SMS messaging. Better capacity and privacy than analog systems.

DATA
Original 2G offers basic circuit data. Later GPRS (2.5G) and EDGE (2.75G) add and improve packet data.

How the network connects

SIMPLIFIED LOGICAL ARCHITECTURE

GSM voice	Digital handset	→ BTS / BSC radio	→ MSC circuit core	→ Telephone network
Later GPRS	GPRS handset	→ BTS / BSC radio	→ SGSN / GGSN	→ Internet

Representative GSM paths shown; IS-95 uses a different CDMA architecture. GPRS / EDGE are later enhancements, not original 2G features.

Advantages

Greater call capacity; SMS; improved privacy and international roaming within compatible systems.

Limitations

Low data rates; limited multimedia; legacy security weaknesses and incompatible regional standards.

Representative use cases

Digital mobile calls, person-to-person texting, basic email and lightweight web access with later data upgrades.

Dates are approximate: commercial rollout differed by country. Capabilities depend on deployment, spectrum, device and coverage; peak rates are not typical user speeds. Navigation controls shown are static.

← Previous · 1G Back to timeline Next · 3G →

Evolution of Mobile Networks · 3G

SET 12 / MOBILE NETWORKS

04 / 07 · ACADEMIC OVERVIEW

3G / EARLY 2000S · APPROXIMATE

3G — Mobile internet becomes practical

Wideband digital access supports richer internet and multimedia services while retaining a separate circuit-switched voice domain.

3G

CORE TECHNOLOGIES

UMTS / WCDMA · CDMA2000
Later: HSPA (UMTS)

VOICE
Usually circuit-switched digital voice; video calling becomes feasible on supported networks and devices.

DATA
Packet-switched mobile internet and multimedia. Later HSPA enhancements improve UMTS data rates and capacity.

How the network connects

SIMPLIFIED LOGICAL ARCHITECTURE

UMTS voice	3G handset	→ Node B / RNC	→ MSC circuit core	→ Telephone network
UMTS data	3G handset	→ Node B / RNC	→ SGSN / GGSN	→ Internet

Simplified UMTS architecture: radio access feeds separate voice and packet-data domains. CDMA2000 follows a different implementation.

Advantages

Higher data capacity than 2G; usable web browsing, email, multimedia and a growing smartphone ecosystem.

Limitations

Variable speeds and coverage; higher network complexity; performance can degrade with congestion or mobility.

Representative use cases

Mobile web and email, photo sharing, navigation, early app services and supported video calls.

Dates are approximate: commercial rollout differed by country. Capabilities depend on deployment, spectrum, device and coverage; peak rates are not typical user speeds. Navigation controls shown are static.

← Previous · 2G Back to timeline Next · 4G →

Evolution of Mobile Networks · 4G

SET 12 / MOBILE NETWORKS

05 / 07 · ACADEMIC OVERVIEW

4G / LATE 2000S–EARLY 2010S · APPROXIMATE

4G — Broadband on an all-IP network

LTE simplifies the packet network and makes high-quality mobile broadband central to everyday digital services.

4G

CORE TECHNOLOGIES

LTE · LTE-Advanced
OFDMA radio · EPC packet core

VOICE
VoLTE carries voice over IP through IMS where supported. Otherwise, devices use legacy circuit-switched fallback.

DATA
All-IP broadband supports high-speed downloads, HD streaming and responsive apps; actual rates vary by conditions.

How the network connects

SIMPLIFIED LOGICAL ARCHITECTURE

Data path	LTE handset	→	eNodeB radio	→	EPC packet core	→	Internet
VoLTE path	VoLTE handset	→	eNodeB + EPC	→	IMS voice service	→	Voice networks

EPC = Evolved Packet Core; IMS = IP Multimedia Subsystem. If VoLTE is unavailable, voice may fall back to a 2G / 3G network.

Advantages

Higher spectral efficiency and throughput; lower latency than 3G; strong support for bandwidth-intensive apps.

Limitations

Coverage and congestion still affect service; voice needs VoLTE support or a legacy fallback network.

Representative use cases

HD video streaming, mobile hotspots, video conferencing, cloud apps and broadband navigation services.

Dates are approximate: commercial rollout differed by country. Capabilities depend on deployment, spectrum, device and coverage; peak rates are not typical user speeds. Navigation controls shown are static.

← Previous · 3G Back to timeline Next · 5G →

Evolution of Mobile Networks · 5G

SET 12 / MOBILE NETWORKS

06 / 07 · ACADEMIC OVERVIEW

5G / 2019 ONWARD · APPROXIMATE

5G — A more adaptable mobile network

5G NR supports enhanced broadband and diverse applications, with capabilities shaped by spectrum and core-network deployment.

5G

CORE TECHNOLOGIES

5G NR · NSA / SA
LTE anchor (NSA) · 5G Core (SA)

VOICE
VoNR on supported standalone deployments; otherwise voice commonly uses VoLTE via LTE fallback or LTE service.

DATA
Enhanced mobile broadband; lower-latency capabilities under suitable conditions; massive device-connectivity goals.

How the network connects

SIMPLIFIED LOGICAL ARCHITECTURE

NSA	5G / LTE handset	→	NR + LTE anchor	→	LTE EPC core	→	Internet / IMS
SA	5G handset	→	gNodeB (NR)	→	5G Core	→	Internet / IMS

NSA = Non-Standalone, using an LTE anchor; SA = Standalone, using a 5G core. VoNR and low-latency services require appropriate network and device support.

Advantages

More flexible capacity and spectrum use; advanced antenna systems; scope for private networks and tailored services.

Limitations

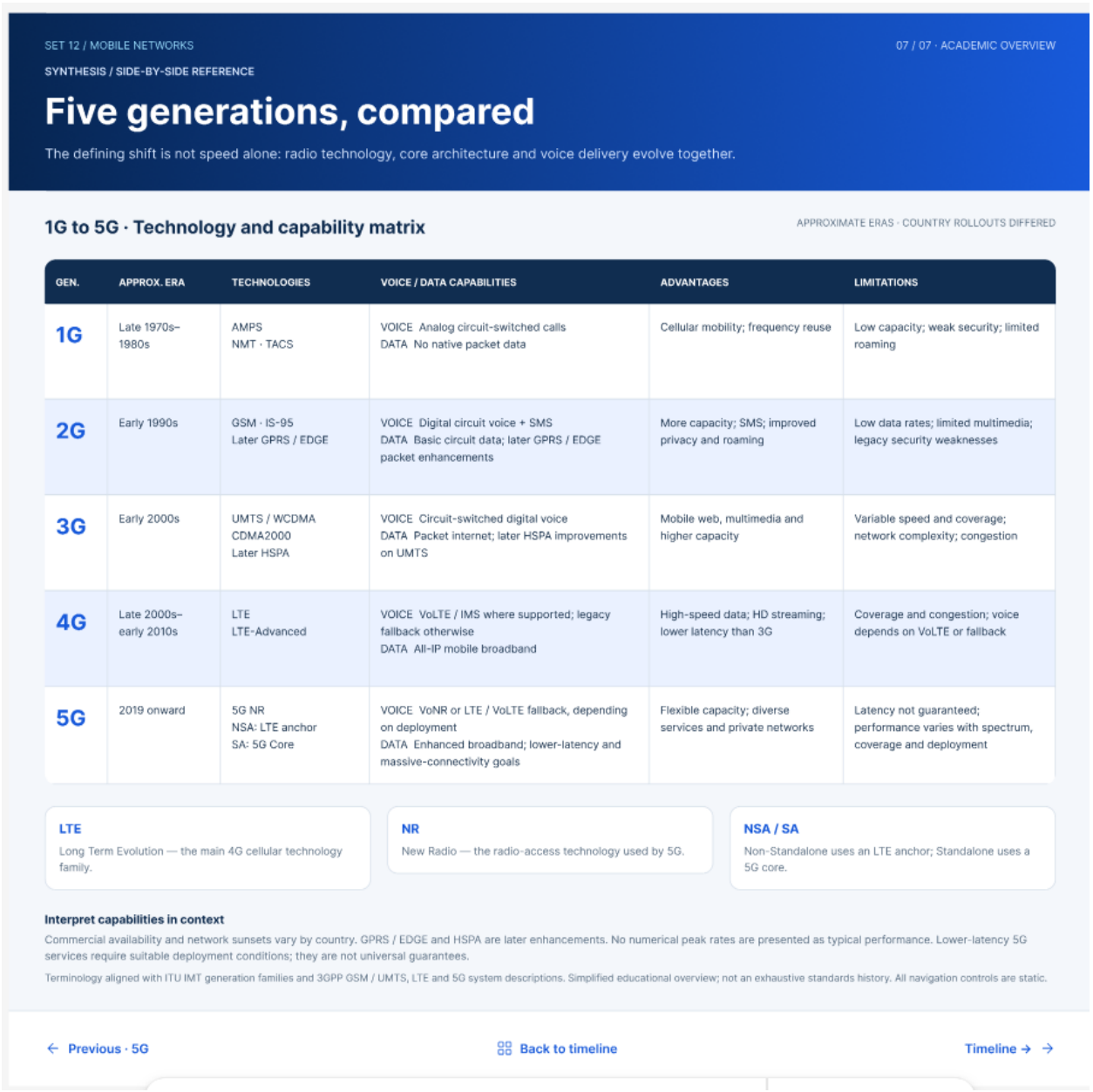
Not all 5G delivers ultra-low latency. Coverage, bands, backhaul, core deployment and devices constrain performance.

Representative use cases

Enhanced broadband, fixed wireless access, industrial private networks, IoT and emerging real-time applications.

Dates are approximate: commercial rollout differed by country. Capabilities depend on deployment, spectrum, device and coverage; peak rates are not typical user speeds. Navigation controls shown are static.

← Previous · 4G Back to timeline Next · Comparison →



Result: -

Hence an interactive diagram illustrating the evolution from 1G to 5G in Figma is created and executed successfully